BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Two-Dimensional Arrays

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How can a loop total an array's values?

2. What is an embedded system designed for?
   A) Any desktop application
   B) Only browsing websites
   C) A specific task
   D) Only storing backups

3. "Adding active users always increases the bandwidth available to each user." Correct this. Explain why.

4. A school needs central accounts and shared files. Three club laptops share files directly. Correct both claims. Explain why. (a) Every client must be a dedicated server. (b) Peer-to-peer needs a central server.

Today's objectives

Store and access two-dimensional array values. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

2D array
Stores data in a fixed-size grid accessed by row and column indices.

Row index
The first index identifies a particular row; indexing starts at zero.

Column index
The second index identifies the column within the chosen row.

Grid assignment
A value is set or retrieved using both its row and column coordinates.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

2D array
Stores data in a fixed-size grid accessed by row and column indices. Apply this in a theatre seats.

Row index
The first index identifies a particular row; indexing starts at zero. Apply this in a school seating plan.

Column index
The second index identifies the column within the chosen row. Apply this in a game board.

Grid assignment
A value is set or retrieved using both its row and column coordinates. Apply this in a class timetable.

I Do - Worked model

A theatre seats is being developed. Explain how 2d array and row index are used. [4 marks]

- Read the scenario and identify the data the program needs.
- 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1].
- Row index [1]; The first index identifies a particular row; indexing starts at zero. [1].
- Two linked points for each concept.

We Do - Guided practice

1. A school seating plan uses 2d array. Explain how it helps. [2 marks]

2. Explain row index in a game board. [2 marks]

3. A class timetable uses column index and grid assignment. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Print the label at row 1, column 0.

```
array desk[2,2]
desk[0,0] = "PC1"
desk[1,0] = "PC2"
```

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how row index and column index help with a museum room matrix. [4 marks]

Q4. A programmer designs a parking spaces. Explain why 2d array, column index and grid assignment matter. [6 marks]

Exit Ticket - True or False?

A two-dimensional array stores data in a grid accessed by row and column indexes.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Set a total to 0. Loop through the array, adding each value once.

2. C - A specific task

3. Users may compete for finite bandwidth, reducing the data rate available to each.

4. (a) A server manages shared services and resources for clients. (b) Peers share directly and can act as clients and servers. Central control suits a school; peer-to-peer can cost less for a small club but is harder to manage as it grows.

I Do - Explain the model

Read the scenario and identify the data the program needs.

2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1].

Row index [1]; The first index identifies a particular row; indexing starts at zero. [1].

Two linked points for each concept.

We Do - Model answers

1. 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1].

2. Row index [1]; The first index identifies a particular row; indexing starts at zero. [1].

3. Column index [1]; The second index identifies the column within the chosen row. [1]. Grid assignment [1]; A value is set or retrieved using both its row and column coordinates. [1].

Code First - Answer

```
array desk[2,2]
desk[0,0] = "PC1"
desk[1,0] = "PC2"
print(desk[1,0])
```

Expected output: PC2

You Do - Self-mark

Q2: 2D array [1]; Column index [1].

Q3: Row index [1]; The first index identifies a particular row; indexing starts at zero. [1]. Column index [1]; The second index identifies the column within the chosen row. [1].

Q4: 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1]. Column index [1]; The second index identifies the column within the chosen row. [1]. Grid assignment [1]; A value is set or retrieved using both its row and column coordinates. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.